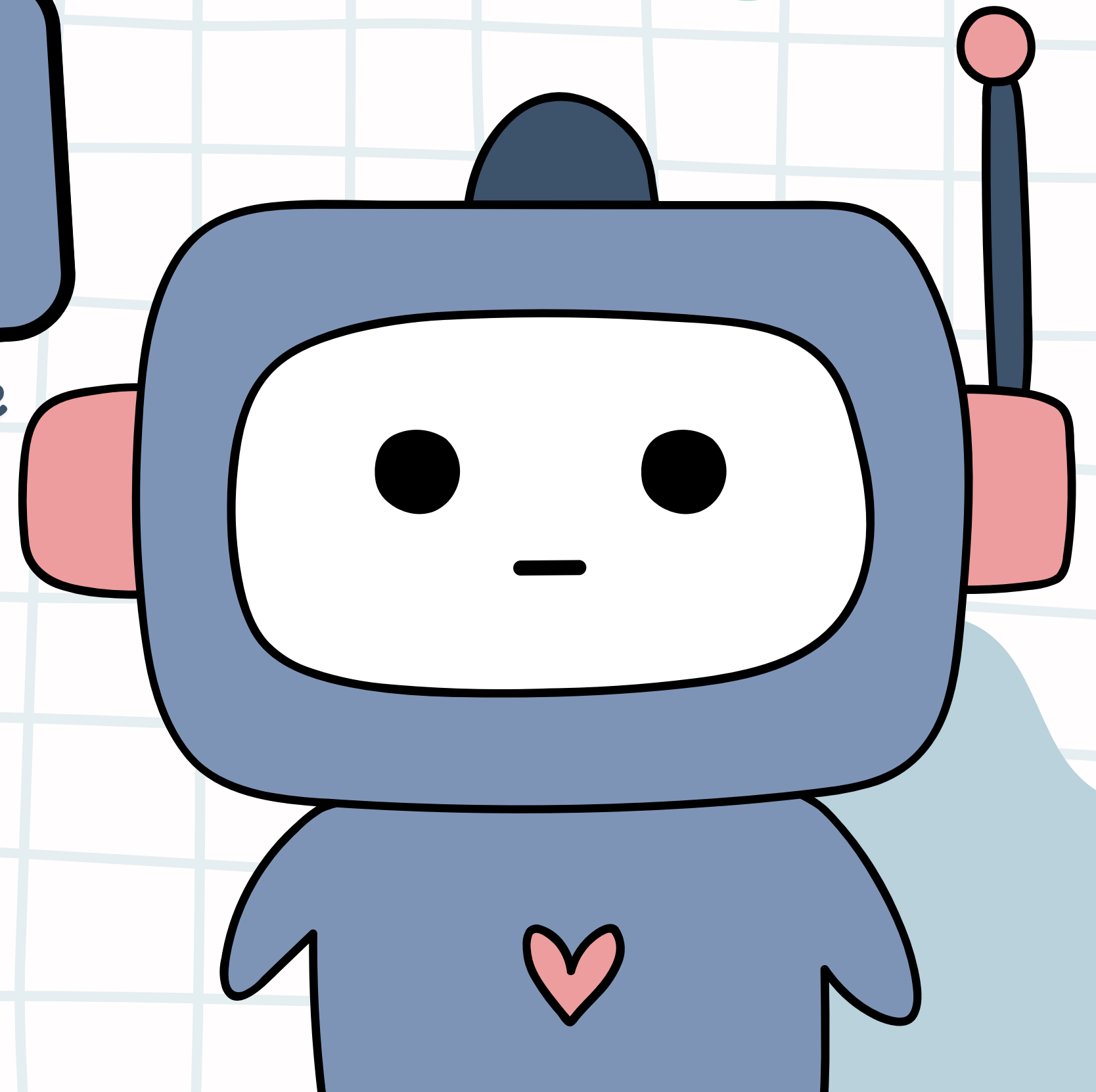


PUZZLE8

An Interactive AI-Based Puzzle Game

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B3_B1 Roll No: 02



8 PUZZLE PROBLEM: AN AI CONCEPT

AI works by exploring possible states and selecting the most efficient path to the goal.

Understanding the Puzzle

- 3×3 grid with 1 blank tile
- Tiles slide to reach the correct order

AI Concept Behind It

- Each board position = a “state”
- Moving a tile = new state
- AI checks different states to reach the goal

Initial State				Goal State		
1	2	3	⇒	1	2	3
8		4		4	5	6
7	6	5		7	8	

WHY INTELLIGENT SEARCH MATTERS?

- 362,880 possible states
- Only ~50% are solvable
- Exploring all states (brute force) is very slow
- Each move creates multiple new possibilities (branching)
- Solution may require 20-30+ steps

Searching blindly wastes time and resources
AI must choose smart paths, not all paths

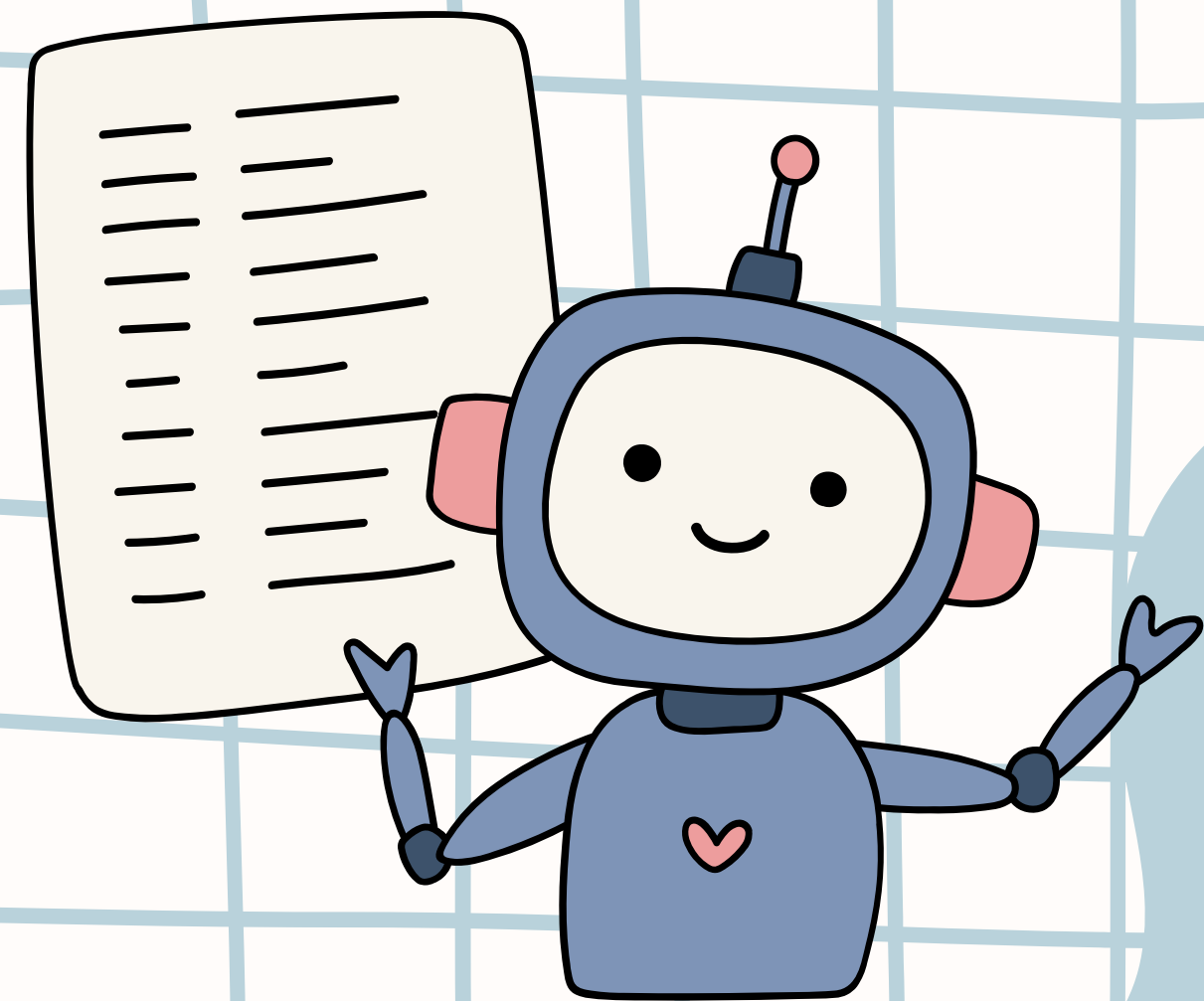
A* SEARCH ALGORITHM

- The Core AI Technique
- A* is an informed search algorithm
- Uses evaluation function:

$$f(n) = g(n) + h(n)$$

- $g(n)$: cost so far
- $h(n)$: estimated cost to goal

Selects most promising state first

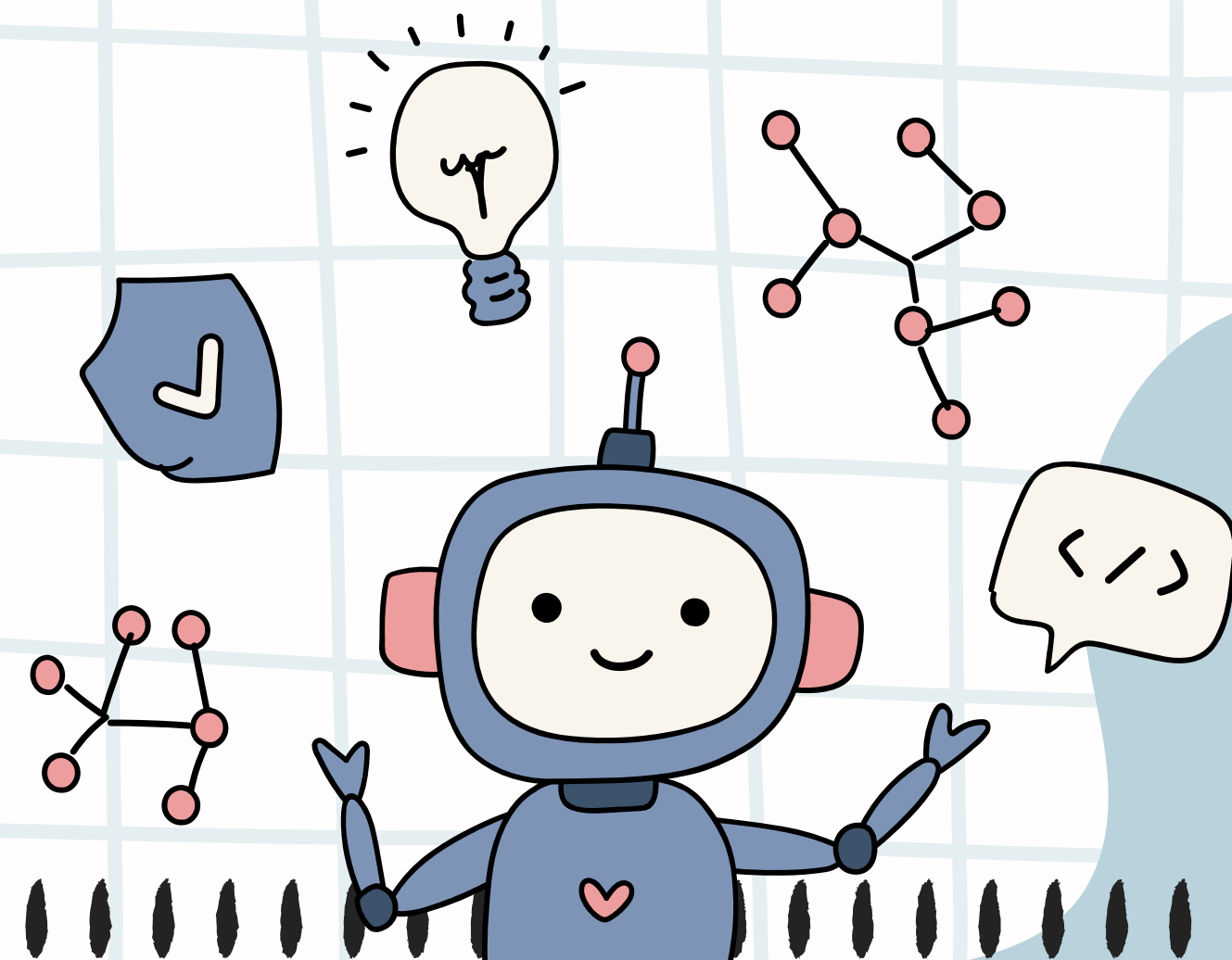


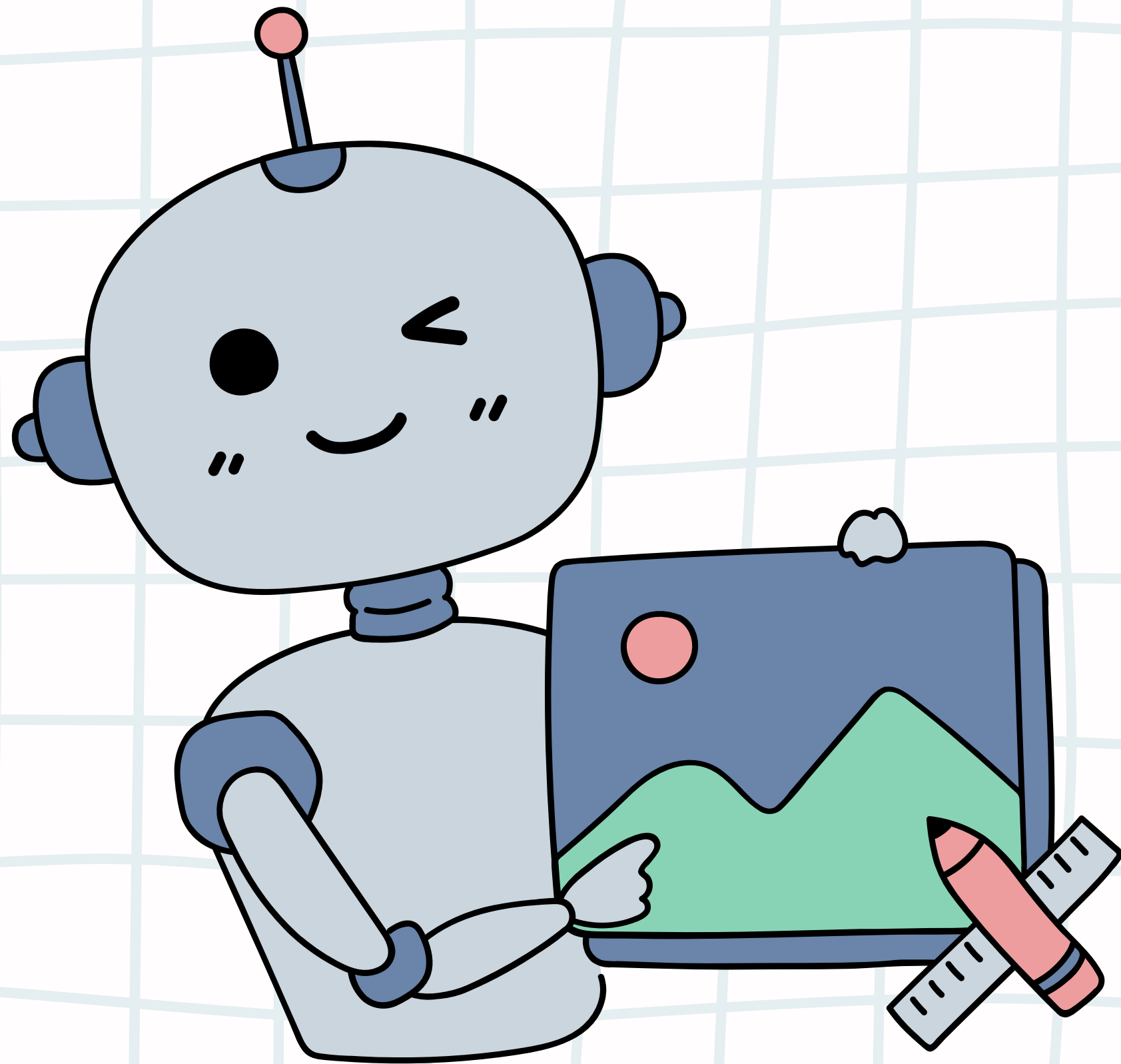
HEURISTICS (AI INTELLIGENCE)

Guiding the Search

- Manhattan Distance → counts how far each tile is from its correct position
- Hamming Distance → counts how many tiles are in the wrong place
- Heuristics estimate “how close we are”

Better heuristic = faster solution

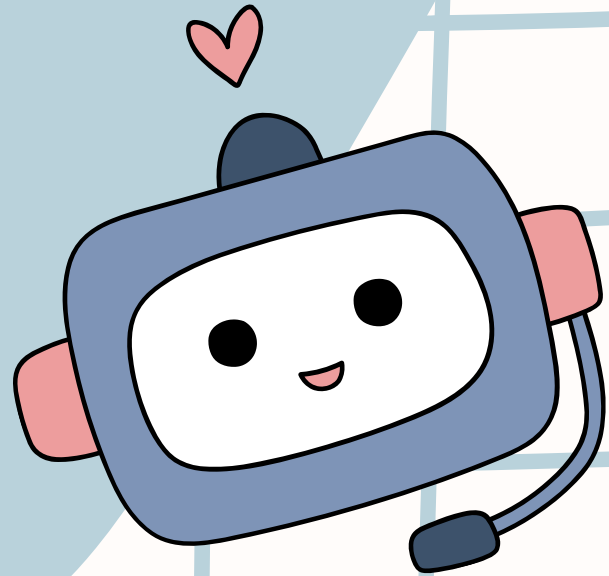




IMPLEMENTATION

System Design

- React + Vite (UI)
 - Tailwind CSS (design)
 - Canvas API (image splitting)
 - LocalStorage (scores)
 - Vercel (deployment & hosting)
- AI logic built separately (modular)



APPLICATION FEATURES

Manual
gameplay

AI auto-solver

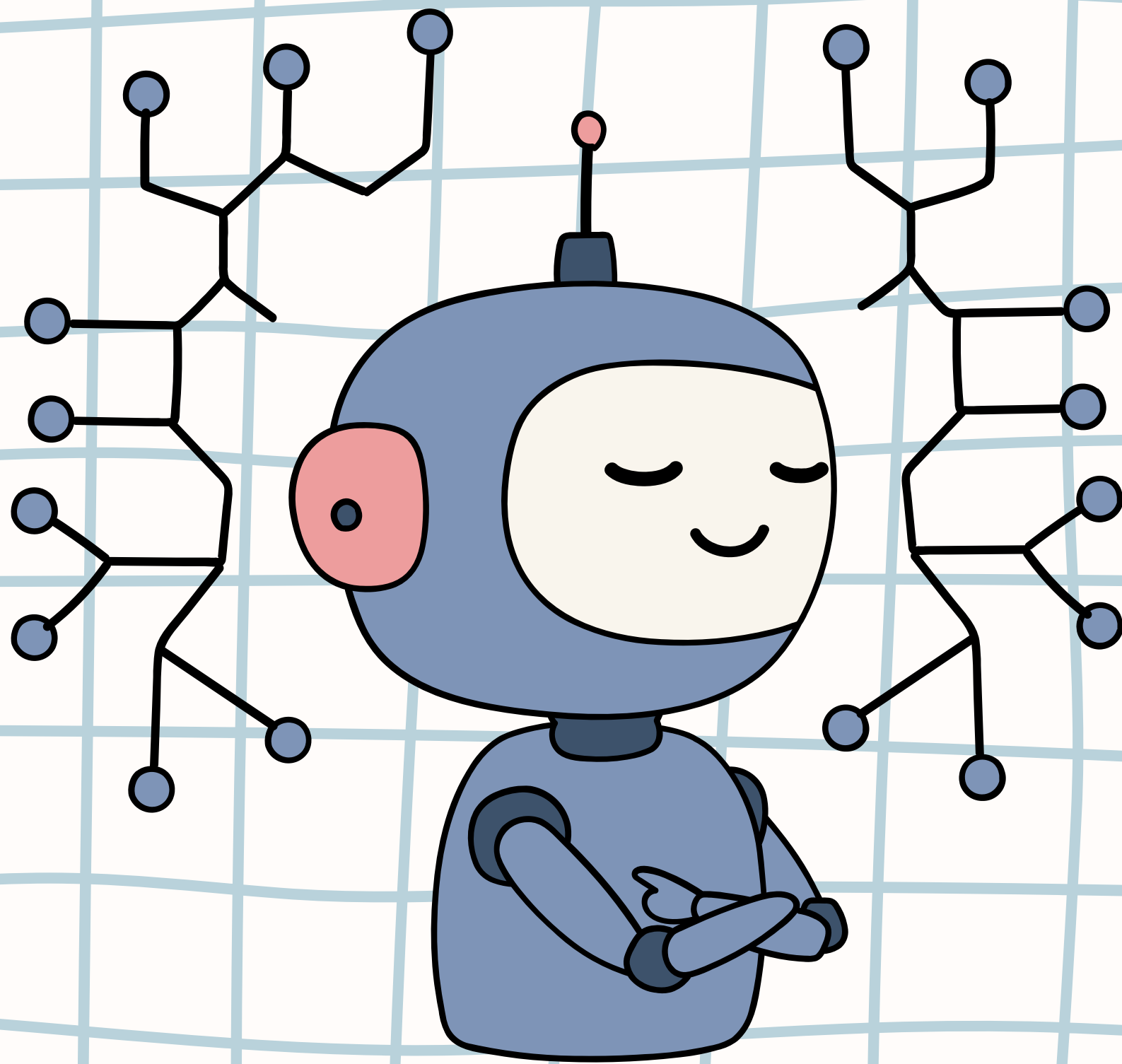
Performance
stats

Image puzzles

Leaderboard

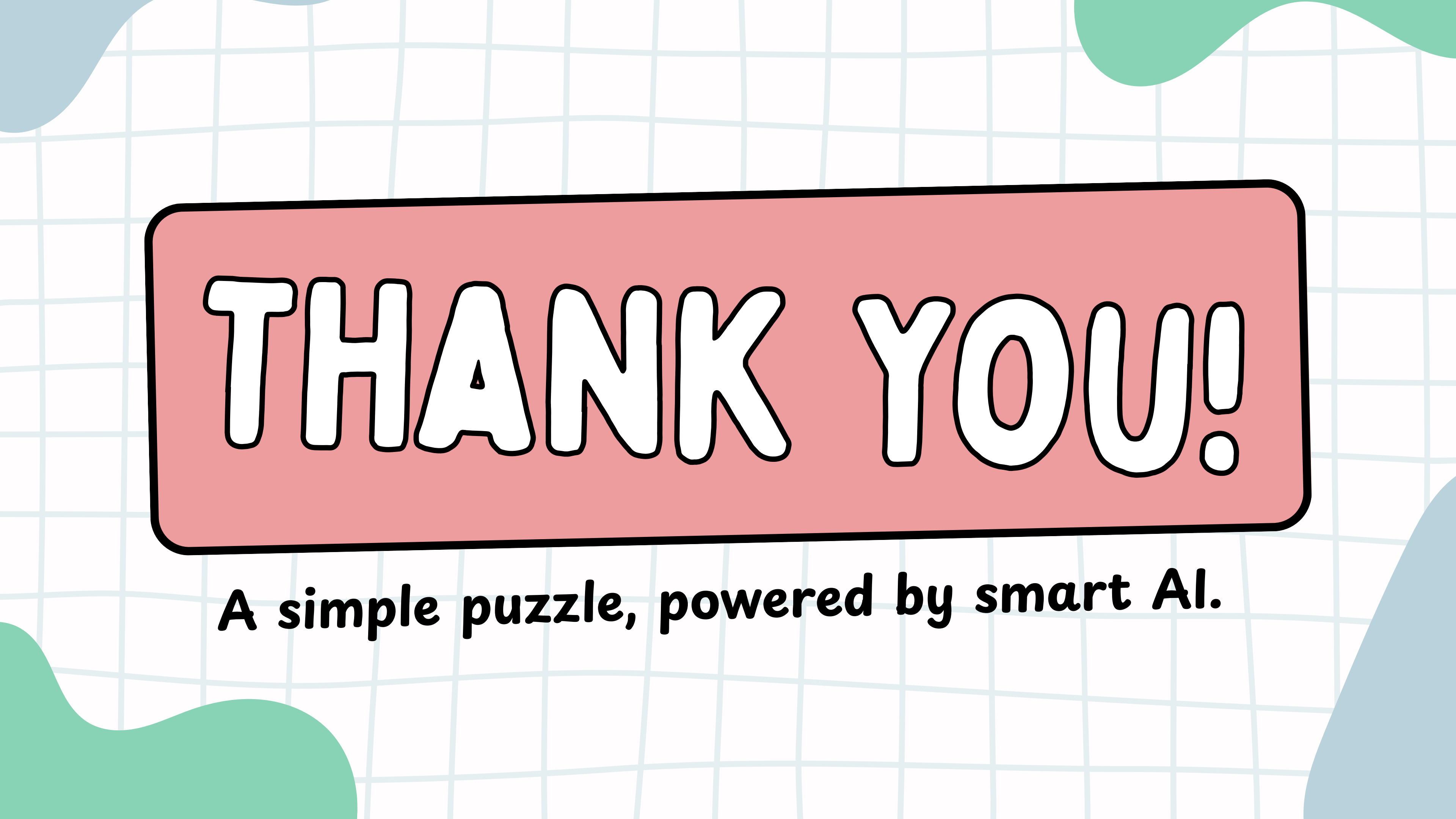
Playtime
Background music

Turns theory into an interactive experience



CONCLUSION

This game implements the A^* algorithm to solve the 8-puzzle efficiently, using intelligent search and heuristics to find the optimal solution and demonstrate how AI selects the best path through smart search techniques.



THANK YOU!

A simple puzzle, powered by smart AI.